# Riparian Zone DEM Generation From Time-Series Sentinel-1 and Corresponding Water Level: A Novel Waterline Method

Jianbo Tan, Mingqiang Chen, Xingyao Xie, Chao Zhang, Beiping Mao, Guangbin Lei, Bo Wang, Xiabing Meng, Xiaobin Guan, and Yunfei Zhang

***Abstract*— Topography data are essential for land management, hydrology, and earth science applications. The topography in a riparian zone varies with time. Available global digital elevation models (GDEMs) do not satisfy the hydrological and environmental modeling requirements for riparian zones. In this study, a novel waterline method was proposed to acquire the digital elevation model (DEM) of riparian zones. Synthetic aperture radar (SAR) images are used to extract waterlines and corresponding water level is applied to acquire the elevation. The uneven water surfaces are considered to acquire the elevation of waterlines and Shuttle Radar Topography Mission (SRTM) DEM is used to interpolate the elevation of pixels without waterlines. The proposed method is applied in the riparian zone of Three Gorges Reservoir and Dongting Lake to test its feasibility. More than 80 Sentinel-1 SAR images at different water levels within three years are used to detect waterlines. Based on corresponding water level of hydrological stations, the elevation of waterlines are acquired via inverse distance weighting method. The distance layer and slope derived from the original SRTM DEM are used to interpolate the elevation of pixels between waterlines. The DEM generated by the proposed method is compared against the DEM derived in 2017, SRTM DEM, the DEM which does not include the uneven effects, and Ice, Cloud and land Elevation Satellite -2 (ICESat-2) ATL08. The good performance of the proposed method implies its high efficiency in revealing riparian zone topography.**

***Index Terms*— Digital elevation model (DEM) generation, riparian zone, time-series synthetic aperture radar (SAR) images, waterline method.**

## I. Introduction

RIPARIAN zones are transitional regions located between aquatic and terrestrial ecosystems. They provide numerous functions and services related to water quality, microclimate regulation, biodiversity, and ecosystem stability [1]. As ecotones, riparian zones play an important role in regulating the materials and energy moving between land and freshwater ecosystems [2]. Riparian zones have garnered a great deal of research attention in regards to watershed ecology and sustainable development [3].

The topography of a riparian zone is one of the main factors that regulates environment and represents essential information in the quantification of water storage and water level fluctuations [4], which is crucial for environmental management and research in riparian zones [5], [6]. A fine and real-time topography data are essential for accurate hydrodynamic simulations in riparian zones [7], [8]. The accuracy of a digital elevation model (DEM) has been shown to have first-order control over the accuracy of hydrodynamic simulations [9].

Global-scale DEMs can be obtained by remote sensing techniques, such as the Shuttle Radar Topography Mission (SRTM) [10], Global Multiresolution Terrain Elevation Data (GMTED2010) [11], Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) global DEM (GDEM) [12], [13] and WorldDEM[1] derived from TerraSAR-X and TanDEM-X missions [14], which provide global topographical data. The SRTM is the most widely used DEM in hydrodynamic simulations [7], [8]. Previous researchers have developed global or nearly global DEMs, but most do not cover riparian zones due to their water fluctuation characteristics [8], [15]. Considering alteration in riparian zone topographies with the driver of land use and cover change (LUCC) and hydrodynamics, former DEMs should be updated to accurately represent current conditions. The InSAR method have been widely applied to update the DEM [16] and contours can be applied to acquire DEM with interpolation method [17]. However, dramatic water-level fluctuations and surface changes prevent their application in riparian zone.

Waterline method has widely been used for reconstructing tidal flat topographies of wide extent regions with the develop-

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ment of time-series satellite [15], [18], [19]. The water surface can be regarded as an altimeter to measure the altitude of a time-series waterline in operating this method [18]. The waterline is defined by the border between the bank and adjacent water region. Waterlines in riparian zones are ever-shifting and can be regarded as a quasi-contour lines of the topography. Riparian zone topography can be determined by a combination of time-series waterlines from different water levels and corresponding water level information [19]. This method efficiently yields DEMs using satellite data [20] and allows for the establishment of riparian DEMs over extensive areas. However, the water surface is uneven due to the inherent characteristics of flowing water along rivers or lakes. The uneven water surface creates different levels of elevation in the waterline and should be incorporated into the riparian zone DEM construction process when operating the waterline method. Waterlines are also irregularly spaced [19], so certain pixels may not be located on the waterline. Appropriate interpolation methods should be applied to predict the values of these pixels.

Synthetic aperture radar (SAR) data can be utilized to retrieve surface water information, as was the case with ENVISAT-advanced SAR (ASAR) and the Sentinel-1 mission [21], [22]. Inundated pixels in the specular reflection of the radar signal on the water surface tend to have low backscattering coefficient and appear dark on SAR [23]. This creates a high land–water contrast and provides useful data for extracting waterlines [24]. Compared to optical satellite images, SAR methods have 24-h and all-weather acquisition capabilities, making them the preferred tool for water mapping [25], [26].

In this study, a novel waterline method incorporating the effects of uneven water surface and void pixels with slope interpolation was developed to acquire riparian zone topographies across the three Gorges reservoir (TGR) and Dongting Lake (DTL). The inverse distance weighting (IDW) method was applied to interpolate the elevation of each point on the waterline [27]. More than 80 Sentinel-1 SAR images covering three years at different water levels were collected to detect the riparian zone waterline. The distance layer and slope derived from the original SRTM DEM were applied to interpolate the elevation of pixels between waterlines. The proposed method was further compared with DEM obtained in 2017, SRTM DEM, a DEM derived by the waterline method not including uneven effects, and Ice, Cloud and land Elevation Satellite -2 (ICESat-2) ATL08 to assess its effectiveness.

The main objectives of this study include establishing a novel waterline method incorporating the effects of uneven water surface and interpolating the elevation of pixels between waterlines with original SRTM DEM and distance layer data.

This article is organized as follows. In Section II, the study area and data are described. In Section III, the method is explained in detail. In Section IV, the results are shown, compared to other datasets and analyzed. In Section V, the merits and limitations of the method are discussed. In Section VI, conclusions of this article are summarized.

## II. Material and Data

### A. Study Area

The riparian zone along the TGR and DTL was selected for case study (Fig. 1). The DTL is located in the middle reaches of the Yangtze River and consists of flood plains. Its mean elevation is approximately 33.5 m, with an average annual temperature of 17 °C and annual rainfall of 1100–1400 mm [28]. The water level data of two hydrological stations (Xiaoheju and Chenglingji) located on either side of the study area is shown in Fig. 1(a).

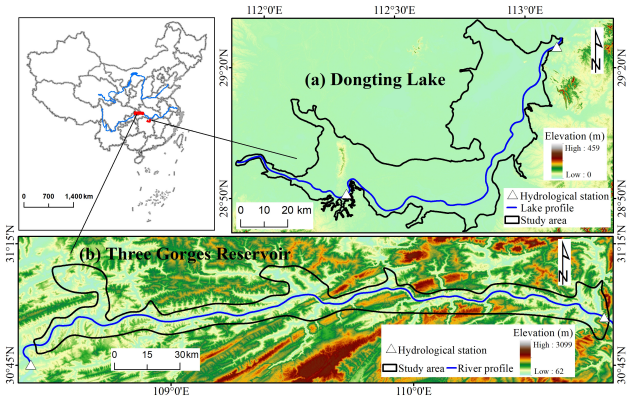


Fig. 1. TGR and DTL locations, hydrological stations, and water profile. (a) DTL. (b) TGR.

The TGR is a geographical region located in the lower section of the upper reaches of the Yangtze River covering the reservoir itself and a surrounding catchment area. Unlike the DTL, its terrain consists of mountains and valleys with elevation ranging from 62 to 3099 m. The case study area is comprised of the main river profile and is dominated by valleys. The area has a humid subtropical monsoon climate with a mean annual precipitation of 1250 mm and mean annual temperature of 16.5 °C [27], [29]. There are two hydrological stations (Wanxian and TGR dam) located on either side of the TGR [Fig. 1(b)].

The TGR riparian zone is narrow and steep and its topography is relatively constant. The riparian zone of the DTL, conversely, is broad and gentle, its topography is more fluctuant due to the effects of hydrodynamics and anthropogenic activities [30]. These two typical riparian zones were isolated to test the effectiveness of the proposed method.

### B. Data Preparation

The Sentinel-1 radar satellite images were used in this study which launched on April 3, 2014, and April 25, 2016. Sentinel-1 is equipped with a C-band (5.405 GHz) SAR, which is a part of the Copernicus program (formerly known as the Global Monitoring for Environment and Security program) [22], [31]. Both Sentinel-1 satellites have near-polar sun-synchronous orbit with an altitude of 693 km and inclination of 98.18°. Their mean local solar time is 18:00 at the ascending node with a 12-day repeat cycle [32]. Both satellites feature-free accessibility, which have expanded the application of Sentinel-1 SAR imagery in the Earth Science field. There are four modes in a Sentinel-1 image including strip map (SM) mode, wave mode (WV), extra-wide (EW)-swath mode and interferometric wide (IW)-swath mode [22].

In this study, the high-resolution ground range detected (GRDH) products acquired in IW mode with VH polarization was used. VH is the kind of polarization for SAR system that the antenna was designed to vertical transmit and horizontal receive waves. There is also more significant contrast

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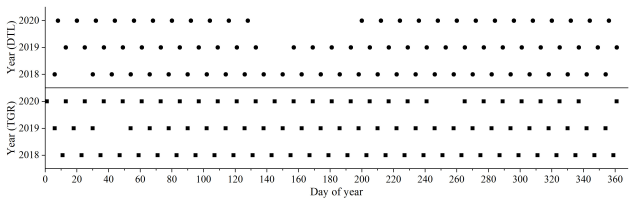


Fig. 2. Sentinel-1 image dates taken in study area.

between water and land in Sentinel-1 VH polarization than VV polarization across the TGR and DTL [33]. The products were downloaded from the Alaska Satellite Facility (ASF). A total of 88 Sentinel-1 images were collected for TGR and 83 images were collected for DTL from 2018 to 2020 (Fig. 2).

Hourly water level data for Wanxian, TGR dam, Xiaoheju, and Chenglingji stations were collected from the ChangJiang Water Resources Commission. Water levels at the time points of Sentinel-1 images were obtained. Considering the local solar time of the Sentinel-1 image and the difference in time zones between TGR and Beijing, the water levels at 19:00 in TGR were obtained to match with the time points in Sentinel-1 images. Water level data for DTL at 18:00 were selected. This two-step selection process maintained consistency in water level time points with the Sentinel-1 images.

The original DEM was obtained from the SRTM at 30 m × 30 m spatial resolution from United States Geological Survey (USGS) [10], then resampled into 10 m size using a nearest sampling method to keep consistent with the pixel spacing of Sentinel-1 images. Slope data were derived from the DEM.

ICESat-2 light detection and ranging (LiDAR) datasets (https://nsidc.org/data/icesat-2/datasets) were applied to further test the accuracy of the DEM derived from the proposed method. The ICESat-2 advanced topographic laser altimetry system (ATLAS) has three laser beams along-track and the cross-track distance between adjacent beams is 3.3 km [34]. The ICESat-2 ATL08, vegetation canopy height, and surface elevation derived from the geolocated point cloud of ATL03 [35], were utilized. ATL08 datasets of DTL from 2018 to 2020 were collected. Only the points of ATL08 uncovered with water were applied for comparison against the corresponding elevation derived via the proposed method.

## III. Methodology

The four-step workflow of the proposed method is shown in Fig. 3 including data preprocessing, water extent and waterline extraction, waterline elevation calculation, and interpolation and validation.

### A. Data Preprocessing

We first preprocessed the Sentinel-1 images to acquire the backscattering coefficient sigma of each pixel. The Sentinel-1 images used in this study are GRDH products in IW mode and VH polarization with a ground resolution of 20 m × 22 m and pixel spacing of 10 m × 10 m in range × azimuth [22]. The data are processed by the sentinel application platform (SNAP) toolkit including thermal noise removal, radiometric calibration, speckle filtering, Doppler terrain correction, and decibel processing [22]. The "Thermal Noise Removal" was applied to remove thermal noise that calculated by the operational

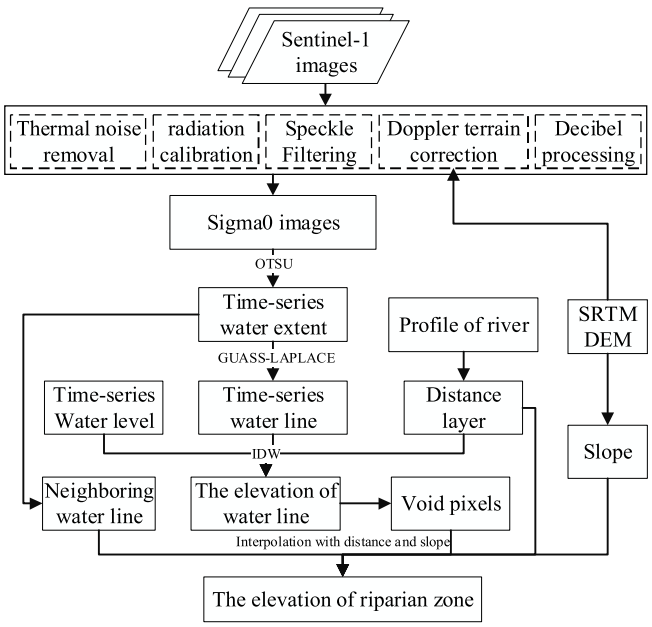


Fig. 3. Workflow of proposed method.

European Space Agency (ESA) instrument processing facility (IPF). Radiometric calibration converts backscatter intensity as received by the sensor to the normalized radar cross section (Sigma0) which is defined in (1) as a calibrated measure taking into account the global incidence angle of the image [36], [37]. Refined Lee filtering was used to remove speckle noise with a 5000.0 edge threshold [22]. The range-Doppler terrain correction operator with 30 m SRTM DEM was used for geocoding and terrain correction [22]. The Sigma0 values were converted into decibel-scale via linear to Sigma naught (decibel) transformation which is presented in (2)

$$\sigma^0 = \mathrm{DN}^2 / A_\sigma^2 \tag{1}$$

$$\sigma_{\mathrm{db}}^0 = 10 * \log_{10}(\mathrm{abs}(f(\sigma^0))) \tag{2}$$

where DN is the digital pixel value; $A_\sigma$ is the sigmaNought derived from look up table (LUT) provided by IPF; $f$ means the processing of Speckle Filtering and Terrain correction; $\sigma_{\mathrm{db}}^0$ is the backscattering image in decibel.

All preprocessing operations were conducted in the SNAP toolbox with the tutorials of Sentinel Toolboxes. The preprocessing workflows were designed using the SNAP graph builder tool, then embedded in the graph processing tool (GPT) to automatically preprocess Sentinel-1 [38].

### B. Water Extent and Waterline Extraction

Open water surfaces have lower sigma0 value than other features due to the specular reflection of radar signals on the water surface for SAR images. Lower sigma0 pixels were placed into water while other pixels were placed into land (Fig. 4). Otsu algorithm is one of the most frequently used automatic thresholding algorithms and based on finding a threshold that minimizes the weighted intraclass variance and maximizes the interclass variance [39], [40]. The Otsu algorithm is relatively simple and easily operated, making it suitable for large SAR image scenes and extracting the time-series water extent [41], [42]. For binary classification, minimizing the variance is equivalent to maximizing

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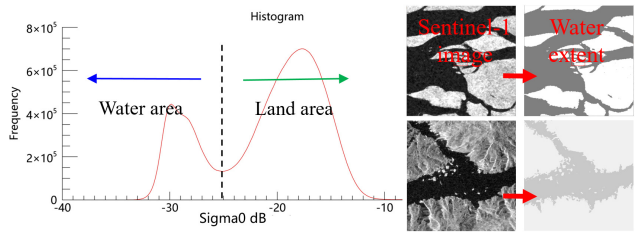


Fig. 4. Sentinel-1 Sigma0 dB histogram and water extent extraction by Otsu algorithm.

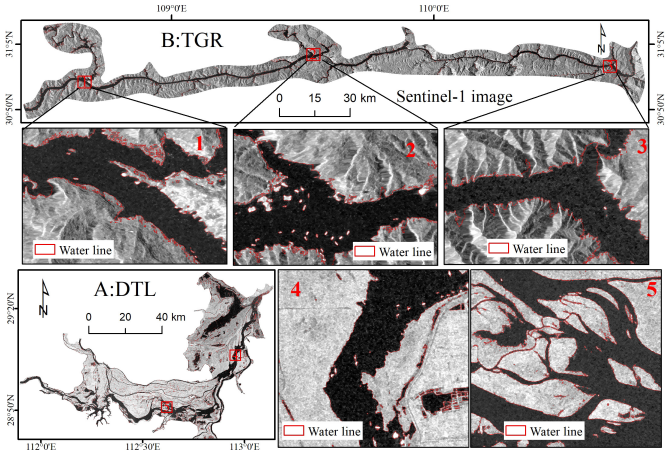


Fig. 5. Waterline obtained by LoG method and local detail of subwindow.

interclass variance. The algorithm exhaustively searches a threshold that minimizes the intraclass variance which is defined in (3). The time-series water extent were extracted based on the histogram thresholding of Sentinel-1 Sigma0_VH values with the Otsu algorithm [39], [43]

$$\mathrm{TH} = \arg\max_t \left(\omega_0(t)\sigma_0^2 + \omega_1(t)\sigma_1^2\right) \tag{3}$$

where TH is the threshold, $\omega_0$ and $\omega_1$ are the probabilities of the two classes separated by a threshold $t$, $\sigma_0$, and $\sigma_1$ are variances of these two classes.

The time-series waterlines were obtained with the Laplacian of Gaussian (LoG) method based on the water extent information. LoG is commonly employed as a second-order edge detector for image processing as it prevents the Laplace from erroneously detecting edges as well as noise [44]. A 2-D LoG was applied in this study to obtain the waterline of the water level (Fig. 5).

### C. Waterline Elevation Calculation

For riparian cross section, the waterline is located at the bottom of the riparian zone when the water level is low, and vice versa (Fig. 6). The elevation of each waterline are equal to the elevation of water surface (water level). The elevation of ground that waterlines laid on are also equal to waterlines' elevation. With water fluctuations, the waterline ranges from the bottom to the top of the riparian zone. Therefore, corresponding water level can be used to obtain the elevation of waterlines ranging from the bottom to the top, ultimately characterizing the elevation in riparian zone. This is the "waterline method" [18].

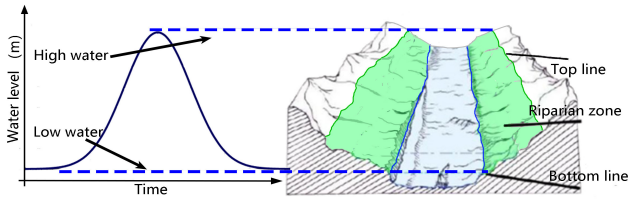


Fig. 6. Illustration of cross section of riparian zone and its water level. The blue and green solid line are bottom and top line.

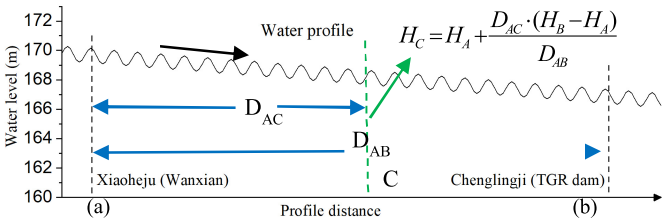


Fig. 7. Water level profile along with the distance. A and B mark locations of hydrological station. C marks the calculated water level position. $H_A$, $H_B$, and $H_C$ are the water levels of A, B, and C, respectively. $D_{AC}$ and $D_{AB}$ are the profile distance between A and C and between A and B, respectively.

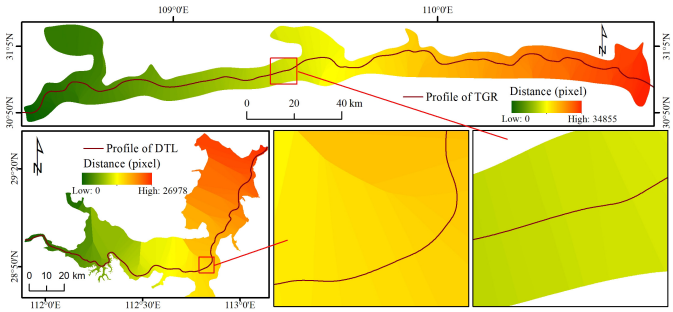


Fig. 8. Water profile distance and local detail of subwindow. The number of pixel is the units of profile distance.

The difference in water level between Wanxian and TGR dam stations ranges from 0.1 to 2.4 m and it ranges from 0.3 to 8.3 m between Xiaoheju and Chenglingji stations in DTL. Therefore, the elevation of the water surface changes with water profile and the same case for the elevation of waterlines (Fig. 7). The uneven effects of water surface should be considered during the application of waterline method. To acquire the elevation of each pixel on waterlines, we assumed that the changes in water levels with water profiles are linearly related and the IDW method is applied to interpolate the elevation of each point on waterlines with the water level of hydrological stations (equation in Fig. 7).

The water profile line in this study is the middle profile line of a river or lake in the dry season. The distance of each pixel on the profile line was first acquired. The distance was the length of a line from the beginning point (Xiaoheju and Wanxian) to its position along with the profile line. The distance of other pixels were assigned with the distance of pixel on the profile line that nearest to them. The profile distances of the study area was presented in Fig. 8. Pixels with same distance forms a transecting cross line which almost perpendicular to the water profile and is referred to here as the "distance line" (Figs. 8 and 9). Coupled with the time-series

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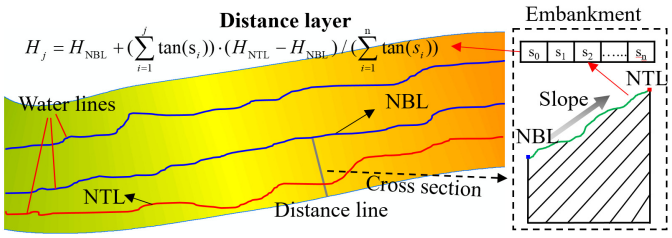


Fig. 9. Interpolating void value with slope. NTL denotes neighboring top line; NBL denotes neighboring bottom line; $H_{NTL}$ and $H_{NBL}$ are the elevation of the NTL and NBL located on the distance line; $H_j$ is the elevation of the $j$th pixel located on the distance line; $S_i$ is the slope value at the $i$th pixel located on the distance line.

water level and the profile distance layer, the water level of waterlines were calculated via IDW method (Fig. 7). There may be more than one waterline located in one pixel and the highest elevation among these waterlines was assigned as the elevation of this pixel.

### *D. Interpolation and Validation*

The elevation of pixels on the waterline was obtained as described above. However, there are some pixels that may never fall on waterlines leading to certain pixels having no elevation after operating the waterline method. The difference in elevation between two positions on embankment transverse line is related to slope accumulation along the route between them which is expressed in (4). Combined with the neighboring waterlines, the slope derived from SRTM DEM on the distance line can be used as a weight to interpolate the elevations of void pixels (Fig. 9). When SRTM DEM is a void pixels, the linear method was applied to interpolate its DEM value.

The neighboring waterline between the top line and void pixel is referred to here as the neighboring top line (NTL). The neighboring waterline between the bottom line and void pixel is the neighboring bottom line (NBL) (Fig. 9). Neighbor pixels located on waterlines and same distance line were used to interpolate the elevation of pixels that do not fall on the waterline with the covariable of slope derived from SRTM DEM. The slope along with distance line, the elevation of NTL ($H_{NTL}$), and NBL ($H_{NBL}$) located on the same distance line were used to acquire the elevation of pixels that do not fall on waterlines with the equation shown in Fig. 9. The interpolation process yielded a gridded DEM of the riparian zone

$$H_n = H_0 + \sum_{i=1}^{n} \tan(\text{slope}_i) \cdot \text{pixel_size} \tag{4}$$

where $H_n$ is the elevation of endpoint, $H_0$ is the elevation of start point, $\text{slope}_i$ is the value of the slope at the $i$th pixel, and pixel_size is the value of spatial resolution of the pixel.

After obtaining a DEM of the riparian zone, the elevation derived from the water level of hydrological stations and water line data in 2017 were applied to further quantitatively estimate the accuracy of the obtained DEM. The $R^2$, mean absolute error (MAE), and root mean squared error (RMSE) between the generated DEM and the elevation in 2017 were calculated to verify the accuracy of the proposed method [27], [45].

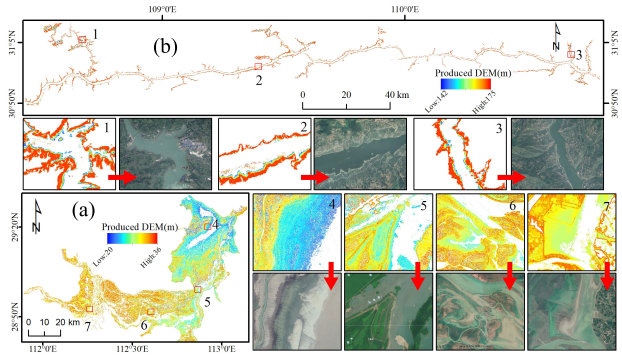


Fig. 10. DEM generated by proposed method, local detail of subwindow and corresponding high-resolution remote sensing image derived from google earth. (a) DTL. (b) TGR.

The elevation between ICESat-2 ATL08 and the generated DEM along with transect lines were compared to test the proposed method. Interpolation via slope was compared to that via linear method to verify its feasibility.

## IV. Result and Analysis

### *A. Generated DEM*

The DEM generated by the proposed method is shown in Fig. 10. Compared with high-resolution remote sensing image derived from Google Earth and Map World in the subwindow, the DEM is closely consistent with the actual topography. The embankment boundaries in DTL are totally represented in the generated DEM and is highly consistent with the actual conditions in the local sites (Fig. 10 subwindows 4–7). The elevation at the beginning of DTL is higher than that at the end and its terrain is relatively flat [Fig. 10(a)]. This is not the case for TGR due to its steep riparian zone and relatively slight difference in water level between Wanxian and TGR dam stations [Fig. 10(b)]. The generated DEM is overall in agreement with the actual topography.

There is more than one water line for one pixels and the numbers of waterlines are shown in Fig. 11. The pixels in steep regions have numerous waterlines and that in plain regions have relatively few (Fig. 11, red circle). For the TGR embankment, the topography of the region at the bottom of the riparian zone is flat and there are few waterlines [Fig. 11(b), subwindow]. The water level in TGR is consistently high which also have contributed to the small number of waterlines at the bottom of the riparian zone. The waterline numbers in riparian zone of TGR is higher than that in DTL which consists of flood plains.

The standard deviations of elevation derived from different waterlines in each pixel are shown in Fig. 12. Regions with fewer waterlines have smaller standard deviations (Figs. 11 and 12). The relatively small sample size may have driven down the variability among them. For TGR, the standard deviation of the region at the bottom of the riparian zone is smaller than at the top [Fig. 12(b), subwindow] which may be attributable to its relatively few waterlines. The high value at the top of the riparian zone implies a large elevation variation among the water lines for same position. The standard deviation is lower overall for DTL, implying a minor variation [Fig. 12(a)].

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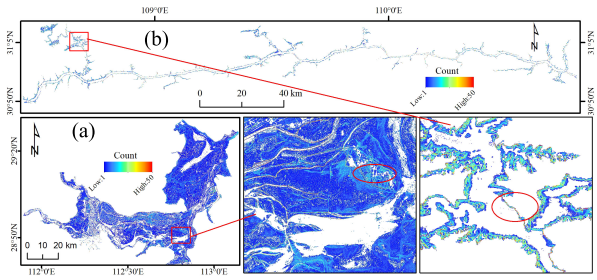


Fig. 11. Map of waterline numbers and local detail of subwindow. Red circle of subwindow in DTL is a village where elevation is high. Red circle in TGR subwindow marks the steep Gaoyang embankment. (a) DTL. (b) TGR.

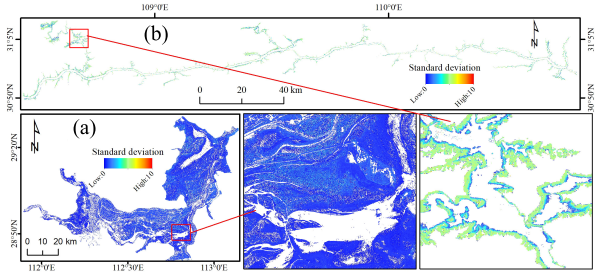


Fig. 12. Standard deviation of water level for each pixel and local detail of subwindow. (a) DTL. (b) TGR.

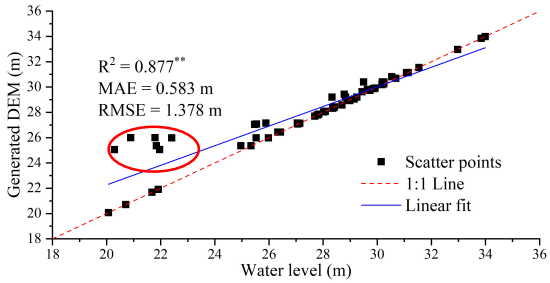


Fig. 13. Scatter plots between generated DEM and elevation derived from water level and water line in 2017, and the $R^2$, MAE. and RMSE between them.

The pixels in the distance line whose profile distance is equal to the profile distance of hydrological station were applied to test the produced DEM. The water level and waterline data of DTL in 2017 was applied for comparison. For the limited number of hydrological sites, only 58 DEM points were applied for validation (Fig. 13). Noise of SAR images or boats in the end of lake or river (Chenglingji station) misclassified water into land which lead to overestimation in elevation with the proposed method (Fig. 13 red circle). The $R^2$ nearly reached to 0.9 and the MAE is 0.583 m, which indicate that the proposed method performed well.

### *B. Comparison Between Generated DEM and SRTM DEM*

Subwindows between the generated DEM and SRTM DEM were compared as shown in Fig. 14. Compared to SRTM DEM, the generated DEM contains a more detailed topography (Fig. 14). For the alterations in the DTL, the SRTM DEM acquired in 2000 differs from the generated DEM in 2018–2020 which is more consistent with the actual topography [Fig. 14(a1)–(a3)]. The generated DEM is highly consistent with the SRTM DEM in TGR, which demonstrates the effectiveness of the proposed method.

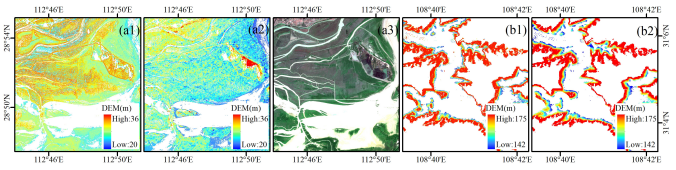


Fig. 14. Produced DEM versus SRTM DEM. (a) Subwindow of DTL. (b) Subwindow of TGR. (a1) and (b1) DEM produced by the proposed method. (a2) and (b2) 30-m SRTM DEM. (a3) True color Sentinel-2 images of subwindow A in 20211108.

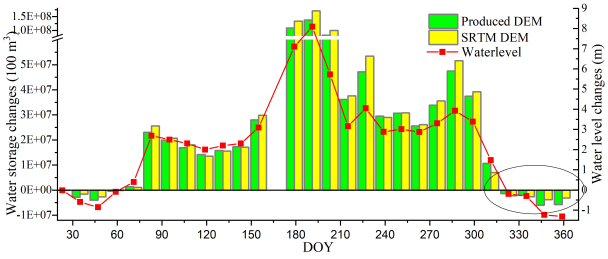


Fig. 15. Comparison among the time-series of water storage changes derived from produced DEM and SRTM DEM in doting lake (the water storage and level changes were calculated by comparing with that in DOY0011).

With the aid of water extent and water level, the time-series of water storage changes in 2017 derived from the generated DEM and SRTM DEM in DTL were compared (Fig. 15). The water storage changes of them are highly consistent with water level changes. In all, the water storage changes derived from SRTM DEM is more significant than that of generated DEM. In the end of 2017 (Fig. 15, circle), water storage changes derived from generated DEM is higher consistent with water level compared to SRTM DEM, indicating a high accuracy of the developed DEM for estimating time series of lake water storage changes.

### *C. Comparison Between Generated DEM and ICESat-2 ATL08*

The transect lines of the ICESat-2 ATL08 of DTL from 2018 to 2020 were applied to validate the generated DEM. The data of four ICESat-2 transect lines in 20181123, 20200107, 20200205, and 20201222 are available in DTL (Fig. 16). When ICESat-2 was overhead, pixels covered with water were removed and leaving 1672 points available for comparison. The footprint size of each photon is 17 m, which is inconsistent with the resolution of the generated DEM [35]. The generated DEM were averaged to keep the data resolution coincident with the 17-m ICESat-2 spot before comparison. The height system of data derived from hydrological stations is the Chinese height datum 1985 (CNHD85) which is different from that of the ICESat-2 ATL08 (ellipsoidal height) [46], resulting in inconsistency between them. However, the scatter plot between ICESat-2 land elevation for the four transect lines are in accordance with the elevation data derived from the generated DEM, the $R^2$ between them is more than 0.65 [Fig. 16(b)]. The residual of the linear regression is relatively low, as are MAE and RMSE [Fig. 16(c)] which imply a close relationship between ICESat-2 ATL08 and the produced DEM.

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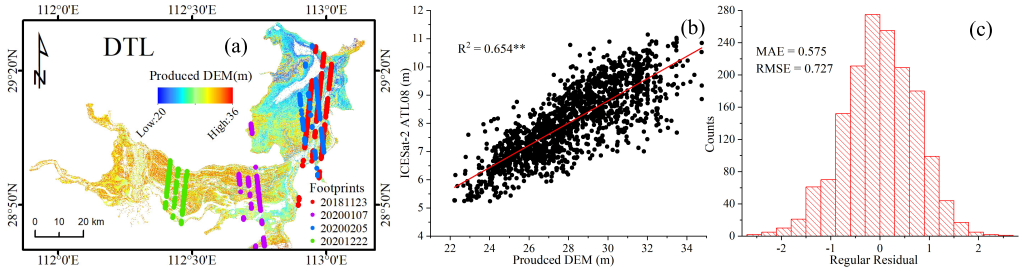


Fig. 16. ICESat-2 ATL08 versus produced DEM. (a) Produced DEM and the location of ATL08 data from 2018 to 2020. (b) Scatter plots of ICESat-2 land elevation and corresponding elevation derived from produced DEM. (c) Histogram of the residual of the linear regression as well as its MAE and RMSE.

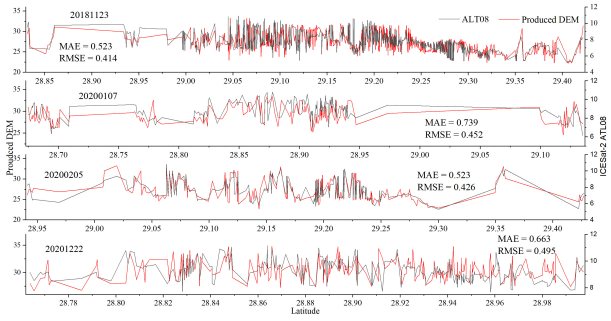


Fig. 17. Topography profiles of the ICESat-2 ATL08 and produced DEM along with transect lines.

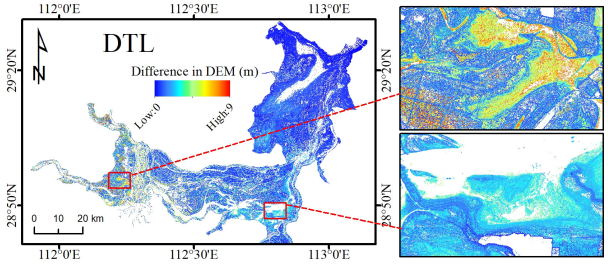


Fig. 18. DEM obtained by proposed method versus DEM produced by waterline method based solely on the water level of Chenglingji station and local detail of subwindow.

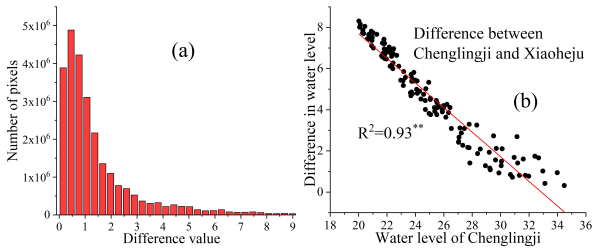


Fig. 19. (a) Difference histogram and scatter plots of Chenglingji water levels. (b) Differences in water level between Chenglingji and Xiaoheju stations.

Fig. 17 shows a comparison between the ICESat-2 ATL08 DEM and the generated DEM along with the transect lines in DTL. There are many differences between them, but the trends of the generated DEM with attitude are consistent with those of the ICESat-2 ATL08 DEM. The MAE is 0.523 and RMSE is 0.414 for the transect line of 20181123, which imply close relationship between ICESat-2 ATL08 DEM and the generated DEM. In the transect line of 20201222 and 20200107, the difference between the generated DEM and ICESat-2 ATL08 DEM is significant compared to others (Fig. 17). These transect lines are located in the region where elevation is high and landscape patterns are more complex (Fig. 16), which may have created uncertainties in the generated DEM.

## V. Discussion

The waterline method is widely applied to extract DEMs. For many inland water regions, however, the uneven water surface and void pixel problems may complicate the traditional waterline method. In this study, we developed a waterline method where the effects of uneven water surface is considered with the IDW method and the void value is interpolated with slope derived from SRTM DEM. The proposed method was applied in DTL and TGR areas to verify its feasibility and effectiveness.

### *A. Effects of Uneven Water Surface*

For many inland water regions, water flow is driven by the difference in water potential. With water potential difference, the water surface is uneven and the attitude of pixels on one waterline are unequal. While operating the waterline method, the waterlines attitude are directly applied to determine the elevation of pixels falling on the waterline. Only applying the water level of one hydrological station or point can thus create uncertainties in the generated DEM. Combining the water level of hydrological stations at the beginning and end of the water profile allows the IDW method to be applied to interpolate the water level of each pixel along the water profile.

The effects of uneven water surface were identified by comparing the DEM generated by the proposed method against a DEM produced by the water level of one hydrological station with the waterline method. The DTL was applied in this study for the sake of comparison. The difference between the DEM generated by the proposed method and a DEM produced by the water level of Chenglingji station with the waterline method is shown in Fig. 18. The difference increases with distance from Chenglingji (Fig. 18) and falls mainly within the range from 0 to 5 m on the histogram [Fig. 19(a)]. The mean value of the difference in DEMs is 1.45 m. Considering that the mean difference in water level between Chenglingji and Xiaoheju station is 2.05 m, the effects of uneven water surface are significant in the DEM generated via waterline method.

The difference among the two methods is significant in areas with low elevation that are far away from Chenglingji station, while it is small in high-elevation areas (Fig. 18 subwindow). With the scatter plots between the water level of Chenglingji and the difference in water level among Chenglingji and Xiaoheju stations, a high water level leads to a slight difference

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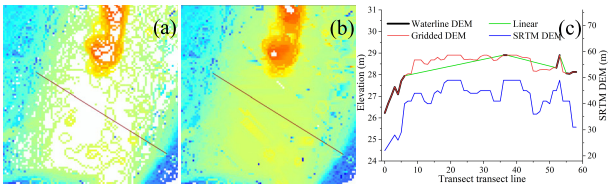


Fig. 20. (a) Waterline DEM of subwindow in DTL. (b) Corresponding gridded DEM with slope-based interpolation. (c) DEM profiles of waterline DEM, gridded DEM, SRTM DEM, and the DEM with linear interpolation along with a transect line [red line in (a) and (b)].

between Chenglingji and Xiaoheju [Fig. 19(b)]. In high-elevation regions, the waterline is distributed only when the water level is high. In this case, the difference in water levels between Chenglingji and Xiaoheju stations is slight and the effects of uneven water surface are insignificant. This may explain why the difference in certain pixels with high elevation and far away from Chenglingji are relatively slight.

In applying the waterline method to generate a DEM, the effects of uneven water surface are significant and are closely correlated with the differences in water levels across different hydrological stations. Differences in water levels across various stations consistently lead to uncertainties in the DEM generated by the waterline method without the consideration of the uneven water surface.

### *B. Void Value Interpolated With Slope Derived From SRTM DEM*

The limitations of satellite images in temporal resolution prevent certain pixels from falling onto the waterline. Void pixels without elevation will emerge during the application of the waterline method. The difference in elevation between two points is equal to the accumulated difference in height of each pixel along with a route between them. Therefore, a slope derived from SRTM DEM can be used to interpolate the void values. SRTM DEM represents a former topography, but the former topography may affect the current slope. To verify the feasibility of slope-based interpolation, we compared this approach to linear interpolation method [19], [22]. Considering the complexity of the waterline DEM in DTL, the slope-based method we used effectively interpolated the void pixels visually [Fig. 20(a) and (b)]. The DEM profile and a transect line of the riparian zone were used to test the performance of the interpolation method with the slope. Compared with linear interpolation, the slope-based interpolation maybe in closer accordance with the actual situation [Fig. 20(c)]. The slope-based interpolation method seems more reasonable.

### *C. Limitations of Proposed Method*

The proposed method effectively generated DEMs in DTL and TGR riparian zones but does have some noteworthy limitations. For example, nearly 100 images were used to extract the waterline and some pixels did not fall properly onto the waterline. With the aid of Google Earth Engine, more satellite data at different water levels (including optical and radar images) could be added to enrich the number of waterlines [22], [47]. The geospatial position accuracy of water extent also directly influences the DEM derived via waterline method. For the preprocessing of SAR images, traditional sigma naught was acquired for classification. However, the terrain-flattened gamma0 normalization can be accounts for topographic variation better than sigma naught [48]. The Otsu method was applied to obtain water extent data. Noise in the SAR images created limitations to the classification method. And in return, misclassifications further affects the waterline results. Especially in mountain region, it may lead to an elevation variation among the water lines for same place (Fig. 12) and create uncertainties in the acquired DEM.

Boats traversing rivers or lakes also introduce uncertainties to the waterline. Water regions containing boats were misclassified as land in our case, which created a number of pixels with high elevation in the middle of river and lake channels—especially in the region around Chenglingji station in DTL (Fig. 10). These erroneous pixels should be removed when generating the DEM and an approach that tolerates these effects should be developed. Furthermore, due to the limited number of hydrological sites, only two hydrological stations laying on the beginning and end of the water profile were applied to acquire the water level of each pixel. The distributary river connecting DTL and TGR may have altered their water levels. More hydrological stations providing such information should be added to improve the precision of water level information for each pixel along the water profile.

The slope derived from the former SRTM DEM was applied to interpolate void values. The former SRTM DEM does not represent the exact slope at present, which may have introduced uncertainties to the gridded DEM. The data resolutions of Sentinel-1 images and SRTM are different. Nine pixels in the Sentinel-1 images were shared with one slope of SRTM DEM, which might cause uncertainties in the process of interpolating the elevation of void pixels between waterlines by the slope derived from SRTM DEM. Furthermore, SRTM did not always map the true ground surface [10]. For the riparian in this study area, the vegetation type are grassland for the stress of water. Even though the signals may easily penetrated into grass, there are some place where the elevation in SRTM does not represent true ground elevation and leading some uncertainties in the interpolation. Considering the lack of latest slope datasets for interpolation, applying the former SRTM DEM is essentially a tradeoff between operation and precision.

Despite these limitations, the proposed method performed well across two large case study areas. The riparian zone of TGR is narrow and steep and that of DTL is broad and gentle, including a dense river net. Strong performance in these two complex areas confirms the feasibility and effectiveness of the ideas in proposed method. Despite limited hydrological station data for some regions, the proposed method can be used to obtain water level information derived from satellite altimetry systems such as Jason, Sentinel-3, and ENVISAT [49], [50] to generate riparian zone DEMs.

## VI. Conclusion

High-resolution topographic mapping of riparian zones was conducted in this study using a novel waterline method based

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on flood-frequency maps derived from Sentinel-1 images and water level data. The uneven water surface was taken into account and the elevation of void pixels were interpolated with slope weight information. The proposed method was applied in DTL and TGR case study areas to test its performance. Despite the complex hydrology of these areas, the proposed method effectively generated DEM which performs comparably to SRTM DEM. The DEM points obtained in 2017, DEM derived via waterline method without the consideration of the uneven effects, and ICESat-2 ATL08 were compared against the DEM derived from the proposed method. The proposed DEM generation method represents precise and recent topographies that can be utilized for hydrology, geomorphology, and ecology applications. An improvements in advanced backscatter estimation methods that incorporate terrain corrections and reduce the noise is expected in the future to achieve a more precision waterlines for the proposed method.

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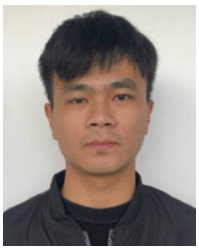

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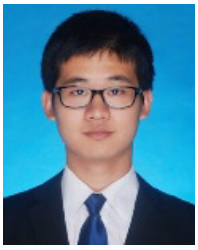

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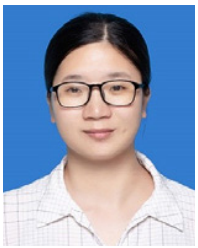

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